

Which Condition Fits? Multi-Label Diagnosis with Supporting Evidence from Symptom Description

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Abstract

Symptom overlap among ADHD, anxiety, and depression makes accurate diagnosis difficult, yet most computational models use single-label classification. This project proposes multi-label text classification with verifiable rationale extraction over self-reported symptom descriptions, highlighting the phrases that support each diagnosis. We review prior approaches grouped by the data they use as evidence (conversational, sensor, clinical, and text-based) and position our project among them. Repository: [click here](#)

Keywords

Mental health, text classification, multi-label classification, rationale extraction

1 Background and Related Work

1.1 Introduction

Several approaches have been used to address this problem, including natural language processing with conversational analysis, multimodal biomarkers with sensor analysis, machine learning on structured clinical data, and text-based analysis of self-reported symptoms. Conversational analysis uses scripted dialogue trees built on diagnostic criteria to screen for disorders during patient conversations [13]. Sensor analysis examines video footage of patients, paying close attention to eye movement and facial expressions, and combines this with audio recordings to detect signs of depression and related distress [1, 4, 8]. Machine learning on structured clinical data applies models ranging from random forests to neural networks to predict the likelihood of a diagnosis and to identify patterns associated with certain mental health conditions [16].

These approaches differ from ours mainly in the data they rely on. Sensor analysis uses video footage and eye tracking, whereas our method uses only text. Conversational analysis asks structured diagnostic questions, whereas we analyze symptom descriptions written in the person’s own words. Structured clinical data approaches draw on a patient’s medical records to support a diagnosis, whereas our method relies only on self-reported text.

All three of these approaches have various strengths and limitations. Conversational agents follow diagnostic criteria but miss milder cases; sensors yield objective markers but rely on small, non-standardized corpora; and clinical-record models scale to large cohorts but raise many false alarms (Section 1.4).

1.2 History

The challenge of classifying mental illness predates computing by well over a century. Nineteenth-century psychiatrists relied on

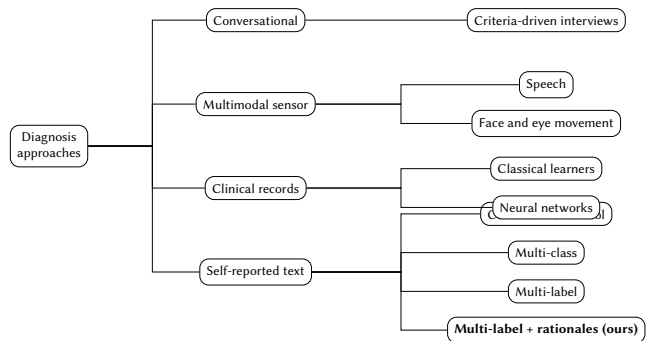


Figure 1: Hierarchy of prior work: input data (level 1), then dialogue design, modality, learner, or label structure (level 2).

inconsistent disease categories until German psychiatrist Emil Kraepelin introduced a more systematic approach to classification in 1899 [9, 10]. His approach distinguished conditions such as dementia praecox from manic-depressive illness based on the course and development of symptoms rather than their surface presentation alone. These developments resurfaced in DSM-III (1980), whose neo-Kraepelinian design replaced psychodynamic categories with explicit operational criteria for each disorder [9]. Language also became a measurable signal: depressed students used more first-person singular pronouns and negative-emotion words in expressive writing [14], and a decade later self-reported diagnoses on Twitter supplied labels for depression and PTSD classifiers [3]; by 2022, social-media posts supplied the data for 81% of 399 reviewed studies [18].

1.3 Hierarchy

We organize prior work by the input data used as diagnostic evidence, which gives four branches (Figure 1): conversational agents, multimodal sensors, structured clinical records, and self-reported text. Sensor work divides by modality (speech versus face and eye movement) and clinical-record work by learner (classical versus neural). Text-based work, the branch closest to ours, divides by label structure: one condition against healthy controls, one of several conditions (multi-class), or several conditions at once (multi-label).

1.4 Comparative Analysis

Philip et al. [13] drove a virtual interviewer with a DSM-5 decision tree, reaching 93% specificity but only 49% sensitivity, lower still for milder cases. Sensor work adds nonverbal signals: DAIC

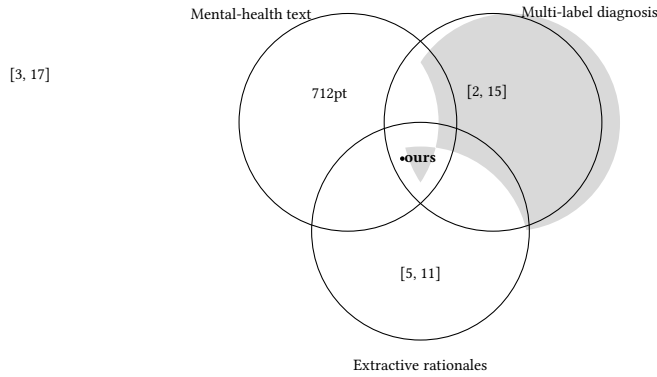


Figure 2: Positioning by three properties (numbers are references). Only our project combines all three; gray regions are outside our scope.

[8] records interview audio and video alongside questionnaire responses, and speech and eye-movement features [1, 4] give objective markers, although corpora are small and collection protocols are not standardized. Clinical-record models scale instead: Tate et al. [16] reached an AUC of 0.74 for 7,638 adolescents from register and parent-questionnaire data, yet a 15% positive predictive value limits screening use.

Text-based methods moved from word counts [14] to neural encoders: a CNN sorted Reddit posts into 11 themes with 71% weighted accuracy [7], and RoBERTa assigned posts to one of five conditions or none [12], but both assign one label per post. Multi-label prediction is rarer: SMHD [2] added a user-level multi-label baseline over nine conditions, and Souza et al. [15] modeled depression-anxiety comorbidity directly (exact-match ratio 0.47), yet their best ensemble dropped the comorbidity classifiers. Explanations developed separately: extractive rationales select the input spans behind a prediction [11] and can be scored for faithfulness and human agreement [5], whereas MentalLaMA [17] generates free-text explanations that need not quote the user’s words. Table 1 summarizes these trade-offs; Figure 2 shows that no reviewed method combines all three properties.

Table 1: Strengths and limitations by category.

Category	Input data	Strengths	Limitations
Conversations	Guided questions in a dialogue	Follows DSM criteria; 93% specificity.	49% sensitivity, lower on milder cases
Sensor	Audio, video, eye movement	Objective, non-invasive markers	Small corpora; no standard protocol
Clinical records	Registers, medical records	Large cohort (AUC 0.74)	15% PPV; no external validation
Text-based	Social-media and forum posts	Abundant natural language	Mostly single-label or vs. controls
Ours	Self-reported symptom text	Multi-label with phrase evidence	Forum labels not clinically verified

1.5 Position Your Project

CLPsych [3] and SMHD [2] are the closest benchmarks, but both classify users from their posting histories, mostly against healthy controls (CLPsych adds one depression-versus-PTSD task), and neither shows the evidence behind a prediction. Because ADHD, anxiety, and depression share symptoms such as inattention and restlessness [6], our project builds on multi-label classification [2, 15] and extractive rationales [11] to separate these commonly confused conditions within a single self-reported description, attaching the supporting phrases to each prediction.

Contributions. Niyaz: history, abstract; Eduardo: introduction, positioning; Thiago: hierarchy, taxonomy; Priyansh: comparative analysis, table.

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